

Two-handed Design: Development of Food Personality Framework Using Mixed Method Needfinding

Summer D. Jung
Stanford University
Stanford, USA
summerjung@stanford.edu

Chandrayee Basu
Stanford University
Stanford, USA
basu.chandrayee@gmail.com

Donghyeon Park
Korea University
Seongbuk-gu, Seoul, South Korea
parkdh@korea.ac.kr

Julie Fukunaga
Stanford University
Stanford, USA
julief@stanford.edu

Maycon Cesar
Tufts University
Medford, Massachusetts, USA
maycon_cesar.santos@tufts.edu

Soh Kim
Stanford University
Stanford, CA, USA
sohyeong@stanford.edu

ABSTRACT

In this case study, we present a mixed methodology approach to needfinding, integrating in-depth qualitative interview data with machine learning-powered analysis of a larger dataset. The research is motivated by the high failure rates and low involvement of consumers in the food startup industry's product design process. To help food startups design products in a more consumer-friendly and timely manner, we are developing a novel framework called the Food Personality Framework (FPF). This framework categorizes eaters according to their eating habits, preferences, motivations and constraints. To better understand the complex relationships between motivations, we chose Grounded Theory as the most pertinent approach and interviewed 14 singles with full autonomy over their food choices. We further leveraged the availability of large online food-related datasets to inform and reinforce our findings from the qualitative work. We analyzed 6687 user behaviors of Food.com, a popular recipe recommendation site, according to the 18 influencers of food choice identified from the qualitative interviews. We found three meaningful clusters of user behavior: Minimalists, Social Butterflies and Conscious eaters. The interview data, thus, enabled grounded classification of the large scale user behavior and provided a grounded way to interpret the relationship among the top motivators identified in each clusters. The cluster analysis will inform future sampling of interviewees and will provide new insightful questions for the qualitative research. The case study delineates a dynamic interplay of qualitative and quantitative data used to investigate human food choice, a novel domain in the Human-Food Interaction literature.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*.

KEYWORDS

Human-Food Interaction, Mixed Method, Automatic Persona Creation, Cluster Analysis, Recipe Data, Grounded Theory, Food Design

ACM Reference Format:

Summer D. Jung, Chandrayee Basu, Donghyeon Park, Julie Fukunaga, Maycon Cesar, and Soh Kim. 2022. Two-handed Design: Development of Food Personality Framework Using Mixed Method Needfinding. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3491101.3503554>

1 INTRODUCTION

1.1 Motivation

For decades, the design literature has emphasized the importance of users [22, 44, 62, 63, 68]. Of many industries, the industry that can particularly benefit from deeper user understanding is food, which is noted for notoriously high failure rates of new products [16, 49]. Human-centered food design is also gaining traction within the Human Computer Interaction community [10, 14, 19]. Because of the relevance of the matter, we conducted exploratory research with the practitioners in the field to investigate the barriers to deeper consumer engagement.

From our interviews with 40 startup founders based in California, USA, we confirmed that the startups used consumers merely as the source of incremental changes to existing products. The commonly reported practices were in contrast with the co-creation model advised by design literature [55, 62, 63]. The lack of resources, including time, labor and knowledge, was found to be the main reason for the deviation from the advised practices. Our findings established the need for better heuristics of consumer understanding. With this in mind, we developed a Food Personality Framework (FPF), to categorize eaters by their influencers of food choice.

1.2 Food Personality Framework

Using personality as an analogy, the Food Personality Framework suggests that different individuals portray different types of eating behaviors, motivated by various factors. Furthermore, we propose that individuals can be grouped based on their most dominant motivations demonstrated in both their words and actions. For instance, some like to prepare meals for an entire week and eat

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3503554>

the same food repeatedly, while others like to explore new foods every day. Each group's common interests and needs around food choices can inform the food companies targeting that group, since marketing messages for each group should be crafted carefully [4, 20, 59]. We differentiate our work from the literature connecting dispositional personality to food behaviors, e.g., the use of Big Five personality types [26] to personalize recipe recommendation [1].

While the Food Personality Framework itself provides valuable information, the current article focuses on the methodological approach used for the development of the framework. Due to the absence of an established theory on the matter, we used Grounded Theory [25] as the appropriate method to develop the theory. During the interviews, the participants often portrayed guilt or shame around their food habits. Acknowledging the potential bias in self-reported data, we complemented the qualitative work with quantitative analysis of large online anonymous behavioral data. Stephens-Davidowitz [69] has pointed out that search queries of anonymous users could be used as a proxy for certain user behaviors, especially the ones that are socially unacceptable. Large online datasets, including those from social media, have been analyzed before to understand various food-related behaviors [18], especially of adolescents [31], and children [71], and in connection with clean and healthy eating [2, 11, 30, 38, 52, 56, 74], and eating disorders [5, 6]. After exploring various publicly available datasets, we chose human recipe interaction data from Food.com. Although the research is still ongoing, the current interplay between the qualitative and quantitative data may benefit the wider Human Food Interaction community.

1.3 Need for Two-handed Design in Consumer Understanding

One of the relevant fields of design for the proposed research is data-driven design, a relatively new concept requiring more research. Although *design process* is a fluid term that encompasses a wide range of practices [34], it consists of a few core stages, which are identifying users' needs, generating ideas, building a prototype, and testing it [35, 58, 72]. The data-driven design literature mostly focuses on using data for brainstorming, prototyping and testing. For instance, Song and Luo [67] developed a data mining algorithm for quicker patent precedent retrieval, which helped the designers to benchmark before brainstorming ideas. Kumar [40] has criticized the limitation of the traditional AB testing and endorsed data mining as a tool to explore design spaces.

Throughout the aforementioned stages of design, identifying users' needs, or needfinding, is mostly left in the realm of qualitative methodology [54], and quantitative data is rarely used to understand users' needs [45]. Quantitative Persona Creation by Salminen et al. [61] is one consolidated field with such an approach. Also referred to as Automatic Persona Generation by Jung et al. [36], this approach allows for the development of consumer persona using quantitative data mined across the Web. Salminen et al. [61] pointed out that the current literature might have missed the deeper, more immersive understanding of users that can only be obtained through qualitative research. In a nutshell, the literature calls for a mixed method approach.

A few recent examples of a mixed method approaches in design include works of [43] and [53]. The former has taken a two-handed approach to understand users' mobile text entry behaviors, analyzing both quantitative data, such as in-situ logs and surveys, and qualitative interviews. Similarly, Park et al. [53] utilized a design workshop and survey to better understand users' needs around email management. While the authors drew meaningful conclusions from the data, they did not attempt to capture the interaction between them. Each dataset functioned as an independent stream of information, not informing each other.

In fact, quantitative and qualitative data have a lot of potential to create synergistic power. On the quantitative side of user research, recommender systems and human-robot collaborative systems have used models of human behavior for predicting user preferences and actions [41, 51, 65, 77]. Quantitative user modeling typically leverages big data and statistical power of Machine Learning. Machine Learning is deductive and starts by organizing (clustering or classification) whatever data is at hand, with an underlying assumption about the data distribution. Examples of using qualitative research for generating behavioral taxonomy is limited in this body of work [3, 29, 64], with the exception of text analysis of social media content [21, 28, 42, 57, 60, 66]. Grounded theory in fact shares similar scientific approach as unsupervised learning, in that they both start with data analysis allowing structures to emerge during the analysis. Like Chen et al. [9], Muller et al. [50], Salminen et al. [61], we observe that Grounded Theory codes are more nuanced and causal, and can be leveraged in other forms of quantitative user analysis, besides just textual analysis. In this case study, we tested a mixed method approach, combining grounded theory and unsupervised learning, to organize users into different buckets of food personality.

2 METHODOLOGY

Qualitative research We started our work with qualitative research, conducting in-depth interviews with participants about their grocery shopping, cooking, and eating behaviors. All participants were pre-screened to be singles in their 20 and 30s and have complete autonomy over their eating decisions. The participants were either recruited through word of mouth or by snowballing [27]. At the end, we interviewed 14 participants including six females, six males, and one non-binary individual. 13 participants submitted their demographics, five of whom identified as White, eight as Asian, one as Black, and one as Native Hawaiian or Pacific Islander. The interviews were conducted using a video conferencing tool due to the public health concerns around COVID-19. The interviews were conversational, and specific incidents were prompted for a deeper understanding of the behaviors [8, 39]. The theme of the interviews ranged from the participants' everyday food behaviors, the food they ate as a child, and their ideal eating experiences. Although each interview was guided by a pre-generated list of questions, the participants were allowed to deviate from the topics for unexpected insights. The interviews lasted for about one hour and were transcribed verbatim for analysis. Following grounded theory literature, we performed open, line by line coding on Atlas.ti [15]. Then, we grouped the low-level codes into larger dimensions of influencers based on their themes. For instance, in Fig. 1, "Got

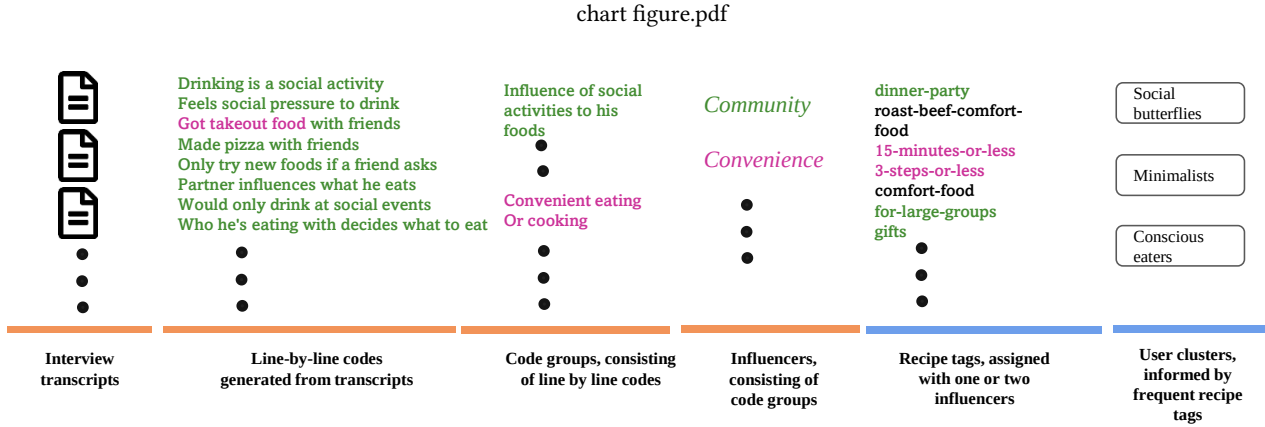


Figure 1: The diagram explains the flow of our mixed method research. The first four steps marked were a part of the qualitative analysis, while the last two steps were quantitative. The green codes altogether form the voluntary motivator "community," and the magenta codes were grouped into "convenience." After converging onto influencers from the Grounded Theory analysis, we used thematic coding to map recipe tags to these influencers. Each user in the recipe data were now represented by a distribution over the influencers and grouped into clusters of different distributions.

takeout food (with friends)" and "Made pizza with friends" were the lowest-level codes derived as summaries of small interview chunks. These codes, along with several other related summaries, were grouped into fewer mid-level codes, which were finally merged to form *Convenience* and *Community*. The themes were further divided into voluntary motivators, involuntary motivators, and desires. All research procedures were approved by the university's Internal Review Board prior to the research.

Behavioral Patterns from a Large Dataset We then analyzed 6687 user behaviors from a popular recipe recommendation site Food.com. We only considered users who had interacted with at least 20 recipes from year 2000 to 2018. The original dataset was contributed by Majumder et al. [46] and had more users. We captured user behavior in terms of the recipes they have interacted with. User engagement ranged from 0 to 6432 days and with 20 to 7671 recipes per user. We also found that on an average 94.7% of the ratings were positive (> 3 out of 5) and 5.28% were negative ratings. Each recipe was pre-annotated with a set of tags that describe its traits, e.g., "15-minutes-or-less," "low-sodium," or "for-large-groups." Since recipe tags are used to enable recipe discovery from the search bar, we used them as features of user behavior. Based on the frequency of the recipe tags, we tested two different clustering schemes. Informed by the findings of the qualitative research, we assumed that user choice is governed by a distribution over the motivators. Each motivator, in turn, could be represented as a combination of tags. This naturally translated into a formulation for topic modeling [75]. We used Latent Dirichlet Allocation with TFIDF features for topic modeling. We observed a power law distribution of tag usage in Food.com, as in several other online recommendation platforms and communities [76]. The most frequent tags dominated the topics and the infrequent tags were non-discriminatory [73], resulting in topics that could not be meaningfully mapped to any influencer. As a second approach, we mapped each tag to one of the 21 highest-level influencers, which were mostly identified in the qualitative

research already. For instance, in Fig. 1, the tags like "dinner-party" and "gifts" map to *Community*, whereas "3-steps-or-less" and "15-minutes-or-less" indicate *Convenience*. Most of the mapping was highly intuitive. We allowed 3 coders to map each tag to multiple influencers, if necessary, then used majority voting to pick a single influencer with the highest probability. We didn't assign any motivator to non-informative search-term tags like dishes, single ingredients or meal types. We used KMeans++ to cluster the users into eater profiles by the frequency of the influencers, normalized by number of interactions for each user. Mapping tags this way resulted in somewhat linear feature distribution, obviating normalization for influencer popularity. To facilitate the exploration of influencer co-occurrences, we developed interactive visualizations of the tags and influencers in a graph form for each cluster. Each node in these graphs is a recipe tag, and the edges represent point-wise mutual information between [12] them.

3 RESULTS

3.1 Motivators Identified

From the interview data, we extracted 18 influencers, consisting of 10 voluntary motivators, three involuntary motivators, and five desires around food. The voluntary motivators were the influencers that positively reinforced certain food choices. For instance, familiarity was one of the elements that the participants looked for when they were deciding what to eat. In contrast, the involuntary motivators functioned as restrictions. The voluntary motivators identified were as follows: community, familiarity, convenience, culture, hedonism, functionality, health, morality, and mood. Overall, the most dominant voluntary motivators were health, hedonism and convenience. Health was closely linked with functionality, both originating from the viewpoint of seeing body as a system and food as input. Although health and hedonism, or seeking sensory pleasure, were both dominant, they showed an inverse relationship. Convenience had multiple components, including easiness

Name	Definition	Example
Community	Eating food to socialize or bond with other people	"In college, I'd always eat with my roommates. Or like, if we're on campus, we'd have the group text like, Hey, we're going to this, wanna meet up? I'm there. Yeah, it was always community focus."
Familiarity	Eating the same food repeatedly or eating what's familiar to them	"I find myself cuisine wise, gravitating towards cultures that have a lot more sour, and they're... [...]so that's like, what citrus vinegar? That stuff that I know I like. "
Convenience	Eating food that is readily available or fast and easy to make	"We bought bread. And then we have like cold cuts lying around the house. And an avocado. That's kind of what I've been eating this week."
Culture	Eating food eaten in one's family or culture	"I'm Japanese. And I'm also part Hawaiian, and it plays just a big role into the types of food I eat [...] So for Japanese culture, I still follow up what my grandma used to do and I have like three layer bento that I bring into office. "
Hedonism	Eating or cooking for fun and pleasure	"I always felt like I had to like spice it up a little bit otherwise that would be like, something that would kind of like, be a little annoying while I'm eating. " "
Functionality	Eating food that serves a specific function	"Then I worked out. Then after that I had a protein shake, like so cuz this thing called Silk protein. It's like, fortified actually."
Health	Eating food that enhances health	"For breakfast, I guess mostly health. [...] I wanted to eat something healthy."
Morality	Eating food that serves moral and political beliefs	"I think that there are some base values like I tried to eat less meat for the environment and because I like animals."
Mood	Eating food that comforts or mentally heals	"The thing of soul food, yeah, like food is like, just like filling for your soul. You know, it could be bad for your body, but as long as your soul feels recharged, you're good to go."
Novelty	Eating novel food out of curiosity	"I think I like trying new things so much that I just don't like planning part of meal planning. Like knowing what you're going to be eating. I'm like, I wanted it to be a surprise!"

Table 1: Voluntary Motivators

Name	Definition	Example
Frugality	Eating food that saves money	"And then takeout, it's kind of expensive about where I live. Um, and so like all the like the delivery fee sometimes, like, if I'm really in a pinch, I'll do like, I'll pick up from like, a local restaurant and and like, I'll kind of divide it so I can eat it like through the week to like be more than one meal."
Physical restriction	Eating food that the body can physically handle	"I'll call the restaurant beforehand to make sure that what I'm probably going to order does not have the things I'm allergic to in it."
Infrastructure	Eating food that is made available by the infrastructure	"When I wasn't at home with like a kitchen and like a solid, like, set of ingredients that I need for like the type of stuff that I make, and my diet and stuff like that, that usually meant like just eating out a lot."

Table 2: Involuntary Motivators

and saving time. The descriptions and examples of the voluntary motivators are presented below in Table. 1.

Furthermore, frugality, or the motivation behind choosing food that saved money, was the most dominant involuntary motivator. Frugality can be a voluntary motivator for certain groups, but for our population of interest, singles in their 20s and 30s, it served

the role of an involuntary motivator. If given money, the participants wanted to spend more on higher quality and more variety of foods. As shown in Table. 2, the other involuntary motivators were physical restrictions, such as allergies, and infrastructure, such as an equipped kitchen. Although we only had one participant with severe food allergies, the impact of the allergies on his life

Name	Definition	Example
Control	Want to have control and transparency	"So I like being able to have a lot more control over like the amount that I eat during the day."
Improvisation	Want to be able to improvise	"One of my goals is to be able to like, improvise in the kitchen and be like, 'okay, I like things this way. And so I'll do it this way.'"
Learning	Want to learn different elements of food and cooking	"If I'm eating alone, I will read or watch videos about the history of what I'm eating."
Arts and crafts	Want to satisfy crafty desires	"I think I like it for the same reason that I really liked doing things like sewing or knitting or drawing[...] that appeal of being kind of like a craft."
Variety	Want to eat a variety of food	"I think my body definitely NEEDS variety. Like I need to have different vegetables, different meats, different proteins, like different, different things like in my body."

Table 3: Desires

was significant. It not only affected his food choices but also made it difficult for him to trust other people, especially waiters, even resulting in psychosomatic issues.

Lastly, the five desires were what the participants looked for in their food-related experiences. The five desires, control, improvisation, learning, art and craft, and variety, are captured in Table 3. In contrast to why the participants chose certain foods, the desires represent how they wanted to eat. For instance, some participants enjoyed the crafty aspect of cooking, but they did not necessarily chose their foods because it yielded crafty experiences. Although they were not the main drivers of the participants' food choices, the desires were mentioned frequently throughout the interviews with a total of 217 instances across 13 out of 14 interviews.

3.2 Trends

On top of the motivators and desires, interesting global trends emerged from the interviews. First, the majority of the participants reported that they did not have as much time and money as they would like to, and there was a stark inverse relationship between time and food, especially healthy food. The time in college was emblematic of such an inverse relationship. The participants would rely on microwavable meals, energy drinks, or fast food. Some reported eating mostly pizza and soda, which have high hedonistic value but lack nutritional diversity. Such choices eventually led to negative physical symptoms, such as gastrointestinal issues or increases in weight. On the other hand, college years were also a formative era, in which the participants started having a better understanding of the role of food. From the negative physical consequences, they realized the need for a diverse food profile, balanced nutrition, and moderation. However, even after college, having a demanding job would make them revert back to unhealthy or convenience-driven eating. Second, almost all participants wished they could eat more quality meals and enhance their relationship with food. When prompted to describe their ideal meals, the participants often shared their desires to be more involved with cooking. They also wanted to explore more variety, spend more time cooking, and have access to more quality ingredients. Furthermore, flavor, a hedonistic value, was more pronounced in the description of ideal

foods. Third, all participants were aware of health, but to a varying degree and with different definitions. While one participant considered her cultural food as healthy food, another regarded plant-based food that one grows and harvests from the ground as healthy food. One common notion of healthy food was nutritionally well-balanced food that includes protein, carbohydrates, vitamins, and so on. Understanding such trends and relationships between the motivators helped us interpret the result of the quantitative analysis as well.

3.3 Recipe Data Analysis

To complement and reinforce qualitative needfinding with a large body of online knowledge, we started analyzing human-recipe interaction data from Food.com. We represented 6687 users by the frequencies of their influencers and clustered them into groups using KMeans++ algorithm. We searched over several possible groupings. We measured cluster quality based on the trade-off between Heterogeneity and Silhouette score, and a good value of Davies-Bouldin Index. Using the above quantitatively, and domain knowledge from qualitative research, we identified an optimal $k = 6$. A low positive Silhouette score indicated overlapping clusters, potentially, due to a continuum of human behavior, observed in many practical applications [32, 47]. In Fig. 2, we present three of the six clusters, of which patterns matched the qualitative data. The *Social butterflies*, who are driven by community, occasion and mood, are less likely to be beginner cooks (low rank of Experience level). In other words, they turn to Food.com when they are cooking for some occasion or a gathering and seldom try out novel healthy recipes. The *Conscious eaters* are health and environment aware, but also prefer palatable dishes (high ranks of Hedonism and Variety). The *Minimalists* are probably beginner cooks in the formative stage of their food autonomy. We also found artifacts of online platform usage in some of the clusters. For instance, some clusters had popular search categories like dishes (e.g., penne pasta) and meal type (e.g., dinner) as the dominant influencers.

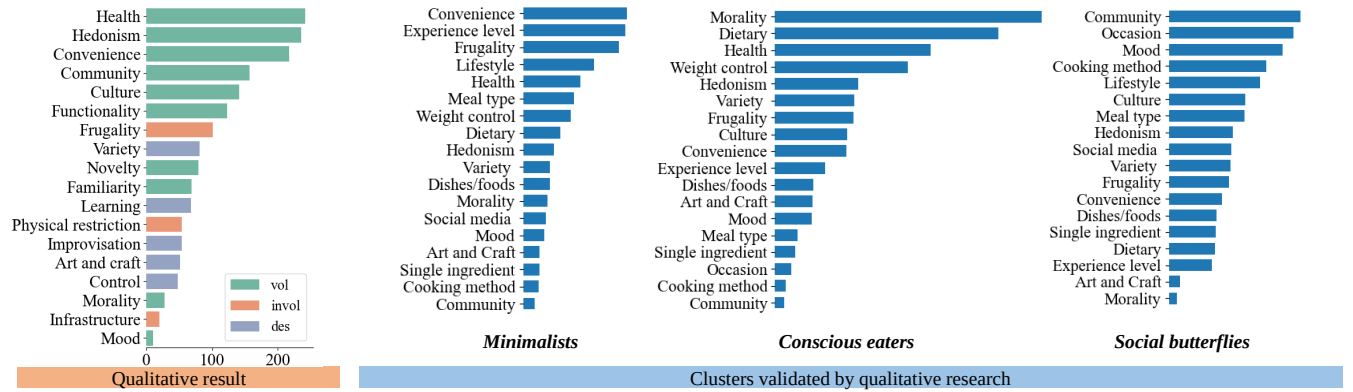


Figure 2: This figure shows the ranked influencers of food choice, obtained from grounded theory coding of open-ended interviews of 14 participants (left) and clustering of recipe interaction data by influencer frequencies from Food.com (right 3). The qualitative data is categorized into voluntary (vol) and involuntary (invol) motivators and desires (des). The three out of six clusters validated by qualitative analysis are named as *Social butterflies*, *Minimalists* and *Conscious eaters*. The x-axis is counts of interview codes for the qualitative drivers and normalized counts of influencers for clustering results.

3.4 Interplay between Qualitative and Quantitative Analysis

From the qualitative data, we extracted the motivators and desires, which we then applied to a large body of online recipe tags. While the recipe tags in themselves displayed power law distribution and hence were non-informative in many cases, assigning influencers to the recipe tags led to meaningful and intuitive user behavior discovery. From the quantitative data, we identified new combinations of motivators and gained new insights about potential relationships between the motivators. The interview data enabled us to conjecture the reasoning behind the clusters formed from the quantitative data. As proof of concept, we found at least one interviewee for each cluster who shared a similar motivation profile.

The *Social butterflies* showed preferences similar to that of our interview participant #8, whose top five motivators were culture, hedonism, community, health, and convenience. She valued the time spent with her family and friends, especially over the holidays. Her favorite memory around food was eating Birria, in Christmas with her family and relatives. *Conscious eaters* showed a similar pattern with participant #17, whose top motivators were health, culture, functionality, convenience, and novelty. Born vegetarian, she decided to continue the diet due to the environmental and ethical concerns. She was also trying to lose weight by low calorie diet. She scored high on culture only because her ethnic food is nutritious, low calorie, and easy to make. From the interview, we were able to see the connection between health, weight loss and plant-based eating. Lastly, the third cluster, *Minimalists*, could be mapped to our participant #14. Her top dimensions were convenience, culture, hedonism, functionality, and health. Being a novice cook, she hoped to be able to improvise and make herself comfortable in the kitchen. She also wanted to spend as little money and time as possible on her food because of two jobs. To her, spending a lot of time for herself felt like "a waste of time." Therefore, she meal prepped, or cooked in bulk and ate the same food repeatedly in a week. Overall, the in-depth interviews with the participants helped us understand

why the clusters were formed in that way. While we were able to match the clusters with the participants in our qualitative research, it is important to note that the groupings of the interview participants are still fluid and adaptable. We will continue to combine and rearrange these personality profiles as we conduct more interviews.

3.5 Interface for exploring motivator co-occurrences

If an individual cares about healthy food, will they prefer quick bites or elaborate meal-kits or eco-friendly packaging? Answers to questions like these can be highly valuable for the practitioners in the food industry, and can be answered if we know how influencers jointly govern an individual's food choices. We found that our recipe interaction data could offer rich information on joint distribution of influencers in an online food community. Our graph interface offered visual exploration of these relationships. The graph represents cluster-specific embeddings of recipe tags, color-coded by influencers. The edge length or distance between the nodes in 2D are inversely proportional to the probability of the co-occurrence of the tags in a given cluster. In Fig. 3, we show two such clusters *Social butterflies* and *Conscious eaters*.

4 DISCUSSION

4.1 Comparison with Existing Literature

In this case study, we present a mixed method approach to understanding food choice drivers, with a focus on the synergistic interplay between qualitative and quantitative methods. The findings from our research are well aligned with the literature and yet add more insights. A large body of literature on Food Choice Questionnaire, of which validity was confirmed in a variety of cultural contexts [17, 33, 48], investigates the motives underlying consumers' food choices [70]. The food choice motivators that the questionnaire considers are as follows: health, mood, convenience, sensory appeal, natural content, price, weight control, familiarity,

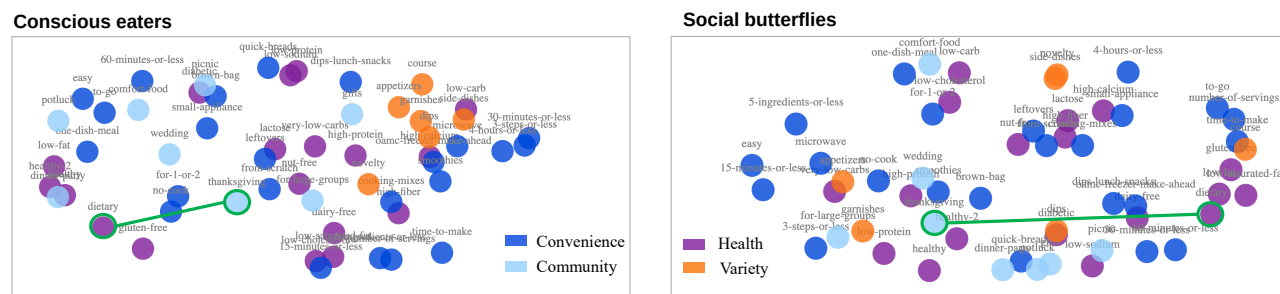


Figure 3: This figure shows two graphs of recipe tags (nodes) color-coded by their influencers for two clusters: *Conscious eaters* and *Social butterflies*. Observe tag-wise and influencer-wise groupings of each cluster. For *Conscious eaters*, *Health* and *Convenience* are grouped together, while *Community* tags are further away. In contrast, *Convenience* was more closely linked to *Community* for *Social butterflies*. The graph shows how relative distances between tags change with eater types, e.g., "dietary" and "thanksgiving" are more likely to co-occur among *Conscious eaters* than among *Social butterflies*, as shown by the green edge between the two nodes.

and ethical concern [24, 70]. Other influencers like novelty and variety are also well documented in the literature. An individual's tendency to seek novelty and familiarity [23] directly affects their food choices during travel. For instance, neophilic people try out foreign food more easily [7, 37]. However, travel experience can also lead to a change in the tendency [13], a phenomena that was observed in our research as well. One interviewee with severe food allergies showed strong adventurousness when he could cook for himself, but the lack of accommodation for his allergies in India led to extreme neophobia. Overall, through triangulation of interview and recipe data, our research shows a wider variety of motivators and adds in-depth insights about the motivators.

5 CONCLUSION AND FUTURE PLAN

As a case study of a mixed method approach to needfinding, this article provides a detailed description of the dynamic interplay between in-depth interviews and larger recipe data analysis. The recipe data, although vast, yield actionable insights for only practitioners in the context of online recipe search. By applying the influencers from the interviews, we were able to discover interpretable clusters of user behavior. We believe that this two-handed approach can be useful not only for human-food interaction community and food industry but also for larger communities in both qualitative and quantitative fields of user research. For instance, recipe recommender systems can also leverage such a mapping from food choices to motivators to lend more interpretability to the system.

One limitation of the current study is the mismatch between the two sub-populations studied, both in terms of demographics and time. The interviewees were singles in their 20s and 30s currently, while the recipe data were generated by users across various ages over a span of 18 years. In the future, we can overcome the issue by finding a targeted human-food interaction dataset or developing a research-specific food-logger. We also plan to leverage the longitudinal nature of the recipe data to identify pivot points in users' food behavior.

The work is still ongoing, and we plan to further develop the framework with more diverse mixed method approaches. First, we will continue interviewing more participants until saturation is achieved and no more motivators and desires are identified. Second, the result of cluster analysis will be utilized to guide the future interviews, in which we will probe into cluster-specific relationships between the influencers. Lastly, we will adopt a multi-dataset approach to quantitative analysis to mitigate any bias in a single dataset. In particular, we are interested in large online human-food interaction datasets that reveal motivators directly, e.g., video logs.

ACKNOWLEDGMENTS

This work was supported by the Stanford Center at the Icheon Global Campus (SCIGC) funded by the Ministry of Trade, Industry, and Energy of the Republic of Korea and managed by the Incheon Free Economic Zone Authority.

REFERENCES

- [1] Ifeoma Adaji, Czarina Sharmaine, Simone Debrowney, Kiemute Oyibo, and Julita Vassileva. 2018. Personality based recipe recommendation using recipe network graphs. In *International Conference on Social Computing and Social Media*. Springer, 161–170.
- [2] Stephanie Alice Baker and Michael James Walsh. 2020. You are what you Instagram: Clean eating and the symbolic representation of food. In *Digital food cultures*. Routledge, 53–67.
- [3] Chandrayee Basu, Qian Yang, David Hungerman, Mukesh Sinhal, and Anca D Drağan. 2017. Do you want your autonomous car to drive like you?. In *2017 12th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 417–425.
- [4] Leydi Caballero, Ana M Moreno, and Ahmed Seffah. 2014. Persona as a tool to involving human in agile methods: contributions from hci and marketing. In *International Conference on Human-Centred Software Engineering*. Springer, 283–290.
- [5] Stevie Chancellor, Tanushree Mitra, and Munmun De Choudhury. 2016. Recovery amid pro-anorexia: Analysis of recovery in social media. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 2111–2123.
- [6] Stevie Chancellor, Jessica Annette Pater, Trustin Clear, Eric Gilbert, and Munmun De Choudhury. 2016. #thyghgap: Instagram content moderation and lexical variation in pro-eating disorder communities. In *Proceedings of the 19th ACM conference on computer-supported cooperative work & social computing*. 1201–1213.
- [7] Richard CY Chang, Jakša Kivela, and Athena HN Mak. 2011. Attributes that influence the evaluation of travel dining experience: When East meets West. *Tourism Management* 32, 2 (2011), 307–316.
- [8] Elizabeth Chell. 1998. Critical incident technique. (1998).

- [9] Nan-Chen Chen, Margaret Drouhard, Rafal Kocielnik, Jina Suh, and Cecilia R Aragon. 2018. Using machine learning to support qualitative coding in social science: Shifting the focus to ambiguity. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 8, 2 (2018), 1–20.
- [10] Jaz Hee-jeong Choi, Conor Linehan, Rob Comber, and John McCarthy. 2012. Food for thought: designing for critical reflection on food practices. In *Proceedings of the Designing Interactive Systems Conference*. 793–794.
- [11] Chia-Fang Chung, Elena Agapie, Jessica Schroeder, Sonali Mishra, James Fogarty, and Sean A Munson. 2017. When personal tracking becomes social: Examining the use of Instagram for healthy eating. In *Proceedings of the 2017 CHI Conference on human factors in computing systems*. 1674–1687.
- [12] Kenneth Ward Church and Patrick Hanks. 1990. Word Association Norms, Mutual Information, and Lexicography. *Computational Linguistics* 16, 1 (1990), 22–29. <https://aclanthology.org/J90-1003>
- [13] Erik Cohen and Nir Avieli. 2004. Food in tourism: Attraction and impediment. *Annals of tourism Research* 31, 4 (2004), 755–778.
- [14] Rob Comber, Eva Ganglbauer, Jaz Hee-jeong Choi, Jettie Hoonhout, Yvonne Rogers, Kenton O'hara, and Julie Maitland. 2012. Food and interaction design: designing for food in everyday life. In *CHI'12 Extended Abstracts on Human Factors in Computing Systems*. 2767–2770.
- [15] Juliet M Corbin and Anselm Strauss. 1990. Grounded theory research: Procedures, canons, and evaluative criteria. *Qualitative sociology* 13, 1 (1990), 3–21.
- [16] C Merle Crawford. 1987. New product failure rates: a reprise. *Research management* 30, 4 (1987), 20–24.
- [17] Luis Miguel Cunha, Diva Cabral, Ana Pinto Moura, and Maria Daniel Vaz de Almeida. 2018. Application of the Food Choice Questionnaire across cultures: Systematic review of cross-cultural and single country studies. *Food quality and preference* 64 (2018), 21–36.
- [18] Munmun De Choudhury, Sanket Sharma, and Emre Kiciman. 2016. Characterizing dietary choices, nutrition, and language in food deserts via social media. In *Proceedings of the 19th acm conference on computer-supported cooperative work & social computing*. 1157–1170.
- [19] Markéta Dolejšová, Rohit Ashok Khot, Hilary Davis, Hasan Shahid Ferdous, and Andrew Quitmeyer. 2018. Designing recipes for digital food futures. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [20] Elina Eriksson, Henrik Artman, and Anna Swartling. 2013. The secret life of a persona: When the personal becomes private. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2677–2686.
- [21] Ethan Fast, Binbin Chen, and Michael S Bernstein. 2016. Empath: Understanding topic signals in large-scale text. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 4647–4657.
- [22] Björn Fischer, Alexander Peine, and Britt Östlund. 2020. The importance of user involvement: a systematic review of involving older users in technology design. *The Gerontologist* 60, 7 (2020), e513–e523.
- [23] Claude Fischler. 1988. Food, self and identity. *Social Science Information* 27, 2 (1988), 275–292. <https://doi.org/10.1177/053901888027002005>
- [24] Christos Fotopoulos, Athanasios Krystallis, Marco Vassallo, and Anastasios Pagiasslis. 2009. Food Choice Questionnaire (FCQ) revisited. Suggestions for the development of an enhanced general food motivation model. *Appetite* 52, 1 (2009), 199–208.
- [25] Barney G Glaser and Anselm L Strauss. 2017. *Discovery of grounded theory: Strategies for qualitative research*. Routledge.
- [26] Lewis R Goldberg. 1990. An alternative" description of personality": the big-five factor structure. *Journal of personality and social psychology* 59, 6 (1990), 1216.
- [27] Leo A Goodman. 1961. Snowball sampling. *The annals of mathematical statistics* (1961), 148–170.
- [28] Margeret Hall, Athanasios Mazarakis, Isabella Peters, Martin Chorley, Simon Caton, Jens-Erik Mai, and Markus Strohmaier. 2016. Following user pathways: Cross platform and mixed methods analysis in social media studies. In *Proceedings of the 2016 CHI conference extended abstracts on human factors in computing systems*. 3400–3407.
- [29] Denise Hebesberger, Tobias Koertner, Christoph Gisinger, Juergen Pripfl, and Christian Dondrup. 2016. Lessons learned from the deployment of a long-term autonomous robot as companion in physical therapy for older adults with dementia a mixed methods study. In *2016 11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 27–34.
- [30] Melanie Hingle, Donella Yoon, Joseph Fowler, Stephen Kobourov, Michael Lee Schneider, Daniel Falk, and Randy Burd. 2013. Collection and visualization of dietary behavior and reasons for eating using Twitter. *Journal of medical Internet research* 15, 6 (2013), e125.
- [31] Christopher Holmberg, John E Chaplin, Thomas Hillman, and Christina Berg. 2016. Adolescents' presentation of food in social media: An explorative study. *Appetite* 99 (2016), 121–129.
- [32] Wasim Huleihel, Arya Mazumdar, Muriel Médard, and Soumyabrata Pal. 2019. Same-cluster querying for overlapping clusters. *arXiv preprint arXiv:1910.12490* (2019).
- [33] Renata Januszewska, Zuzanna Pieniak, and Wim Verbeke. 2011. Food choice questionnaire revisited in four countries. Does it still measure the same? *Appetite* 57, 1 (2011), 94–98.
- [34] Ulla Johansson-Sköldberg, Jill Woodilla, and Mehves Çetinkaya. 2013. Design thinking: past, present and possible futures. *Creativity and innovation management* 22, 2 (2013), 121–146.
- [35] Robert Johnsey. 1995. The design process—Does it exist? *International Journal of Technology and Design Education* 5, 3 (1995), 199–217.
- [36] Soon-gyo Jung, Joni Salminen, Haewoon Kwak, Jisun An, and Bernard J Jansen. 2018. Automatic Persona Generation (APG) A Rationale and Demonstration. In *Proceedings of the 2018 conference on human information interaction & retrieval*. 321–324.
- [37] Yeong Gug Kim, Anita Eves, and Caroline Scarles. 2009. Building a model of local food consumption on trips and holidays: A grounded theory approach. *International journal of hospitality management* 28, 3 (2009), 423–431.
- [38] Karen Michelle Klassen, Emily S Borleis, Linda Brennan, Mike Reid, Tracy A McCaffrey, and Megan SC Lim. 2018. What people "like": Analysis of social media strategies used by food industry brands, lifestyle brands, and health promotion organizations on Facebook and Instagram. *Journal of medical Internet research* 20, 6 (2018), e10227.
- [39] Gary A Klein, Roberta Calderwood, and Donald Macgregor. 1989. Critical decision method for eliciting knowledge. *IEEE Transactions on systems, man, and cybernetics* 19, 3 (1989), 462–472.
- [40] Ranjitha Kumar. 2019. Data-Driven Design: Beyond A/B Testing. In *Proceedings of the 2019 Conference on Human Information Interaction and Retrieval*. 1–2.
- [41] Nicholas D Lane, Ye Xu, Hong Lu, Andrew T Campbell, Tanzeem Choudhury, and Shane B Eisenman. 2011. Exploiting social networks for large-scale human behavior modeling. *IEEE Pervasive Computing* 10, 4 (2011), 45–53.
- [42] Zachary Levonian, Drew Richard Erikson, Wenqi Luo, Saumik Narayanan, Sabirat Rubya, Prateek Vachher, Loren Terveen, and Svetlana Yarosh. 2020. Bridging qualitative and quantitative methods for user modeling: Tracing cancer patient behavior in an online health community. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 14. 405–416.
- [43] Toby Jia-Jun Li and Brad A Myers. 2021. A Need-finding Study for Understanding Text Entry in SmartphoneApp Usage. *arXiv preprint arXiv:2105.10127* (2021).
- [44] Yuanhua Liu, Anna-Lisa Osvalder, and MariAnne Karlsson. 2010. Considering the importance of user profiles in interface design. In *User interfaces*. IntechOpen.
- [45] Jianxi Luo, Bowen Yan, and Kristin Wood. 2017. Innogps for data-driven exploration of design opportunities and directions: The case of google driverless car project. *Journal of Mechanical Design* 139, 11 (2017).
- [46] Bodhisattwa Prasad Majumder, Shuyang Li, Jianmo Ni, and Julian McAuley. 2019. Generating personalized recipes from historical user preferences. *arXiv preprint arXiv:1909.00105* (2019).
- [47] Xueyu Mao, Purnamrita Sarkar, and Deepayan Chakrabarti. 2018. Overlapping clustering models, and one (class) svm to bind them all. *arXiv preprint arXiv:1806.06945* (2018).
- [48] Jerko Markovina, Barbara J Stewart-Knox, Audrey Rankin, Mike Gibney, Maria Daniel V de Almeida, Arnout Fischer, Sharron A Kuznesof, Rui Póinhos, Luca Panzone, and Lynn J Frewer. 2015. Food4Me study: Validity and reliability of Food Choice Questionnaire in 9 European countries. *Food quality and preference* 45 (2015), 26–32.
- [49] Chris Muller and Robert H Woods. 1991. The real failure rate of restaurants. *Hospitality Review* 9, 2 (1991), 7.
- [50] Michael Muller, Shion Guha, Eric PS Baumer, David Mimno, and N Sadat Shami. 2016. Machine learning and grounded theory method: convergence, divergence, and combination. In *Proceedings of the 19th international conference on supporting group work*. 3–8.
- [51] Tatsuya Nomura, Takayuki Kanda, Tomohiro Suzuki, and Kensuke Kato. 2008. Prediction of human behavior in human–robot interaction using psychological scales for anxiety and negative attitudes toward robots. *IEEE transactions on robotics* 24, 2 (2008), 442–451.
- [52] Ferda Ofli, Yusuf Aytar, Ingmar Weber, Raggi Al Hammouri, and Antonio Torralba. 2017. Is saki# delicious? the food perception gap on instagram and its relation to health. In *Proceedings of the 26th International Conference on World Wide Web*. 509–518.
- [53] Soya Park, Amy X Zhang, Luke S Murray, and David R Karger. 2019. Opportunities for automating email processing: A need-finding study. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [54] Dev Patnaik and Robert Becker. 1999. Needfinding: the why and how of uncovering people's needs. *Design Management Journal (Former Series)* 10, 2 (1999), 37–43.
- [55] Erika Maria Perttunen, Summer D Jung, Senni Kirjavainen, Tua A Björklund, and Sohyeong Kim. 2021. THE ROLE OF CONSUMER INFLUENCES IN THE NEW PRODUCT DEVELOPMENT OF PACKAGED FOOD AND BEVERAGE VENTURES. *Proceedings of the Design Society* 1 (2021), 3111–3120.
- [56] Ladislav Pilař, Lucie Kvasničková Stanislavská, and Roman Kvasnička. 2021. Healthy food on the twitter social network: Vegan, homemade, and organic food. *International Journal of Environmental Research and Public Health* 18, 7 (2021),

- 3815.
- [57] Claudette Pretorius, Darragh McCashin, Naoise Kavanagh, and David Coyle. 2020. Searching for mental health: a mixed-methods study of young people's online help-seeking. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [58] Rim Razzouk and Valerie Shute. 2012. What is design thinking and why is it important? *Review of educational research* 82, 3 (2012), 330–348.
- [59] Adele Revella. 2015. *Buyer personas: how to gain insight into your customer's expectations, align your marketing strategies, and win more business*. John Wiley & Sons.
- [60] Joni Salminen, Hind Almerikhi, Milica Milenković, Soon-gyo Jung, Jisun An, Haewoon Kwak, and Bernard J Jansen. 2018. Anatomy of online hate: developing a taxonomy and machine learning models for identifying and classifying hate in online news media. In *Twelfth International AAAI Conference on Web and Social Media*.
- [61] Joni Salminen, Kathleen Guan, Soon-gyo Jung, Shammur A Chowdhury, and Bernard J Jansen. 2020. A literature review of quantitative persona creation. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [62] Elizabeth B-N Sanders. 2002. From user-centered to participatory design approaches. In *Design and the social sciences*. CRC Press, 18–25.
- [63] Elizabeth B-N Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *Co-design* 4, 1 (2008), 5–18.
- [64] Johanna Seibt, Christina Vestergaard, and Malene F Damholdt. 2021. The Complexity of Human Social Interactions Calls for Mixed Methods in HRI: Comment on "A Primer for Conducting Experiments in Human-robot Interaction," by G. Hoffman and X. Zhao. *ACM Transactions on Human-Robot Interaction (THRI)* 10, 1 (2021), 1–4.
- [65] Subhash K Shinde and Uday Kulkarni. 2012. Hybrid personalized recommender system using centering-bunching based clustering algorithm. *Expert Systems with Applications* 39, 1 (2012), 1381–1387.
- [66] Philipp Singer, Florian Lemmerich, Robert West, Leila Zia, Ellery Wulczyn, Markus Strohmaier, and Jure Leskovec. 2017. Why we read Wikipedia. In *Proceedings of the 26th international conference on world wide web*. 1591–1600.
- [67] Binyang Song and Jianxi Luo. 2017. Mining patent precedents for data-driven design: the case of spherical rolling robots. *Journal of Mechanical Design* 139, 11 (2017).
- [68] Marc Steen, Menno Manschot, and Nicole De Koning. 2011. Benefits of co-design in service design projects. *International Journal of Design* 5, 2 (2011).
- [69] Seth I Stephens-Davidowitz. 2013. The cost of racial animus on a black presidential candidate: Using google search data to find what surveys miss. *Available at SSRN* 2238851 (2013).
- [70] Andrew Steptoe, Tessa M Pollard, and Jane Wardle. 1995. Development of a measure of the motives underlying the selection of food: the food choice questionnaire. *Appetite* 25, 3 (1995), 267–284.
- [71] LeeAnn Tan, See Hoe Ng, Azahadi Omar, and Tilakavathi Karupaiah. 2018. What's on YouTube? A case study on food and beverage advertising in videos targeted at children on social media. *Childhood Obesity* 14, 5 (2018), 280–290.
- [72] Katja Tschimmel. 2012. Design Thinking as an effective Toolkit for Innovation. In *ISPIM Conference Proceedings*. The International Society for Professional Innovation Management (ISPIM), 1.
- [73] Conor Hayes Paolo Avesani Sriharsha Veeramachaneni, C Hayes, and P Avesani. 2007. An analysis of the use of tags in a blog recommender system. In *IJCAI'07*. 2772–2777.
- [74] Leticia Vidal, Gastón Ares, Leandro Machín, and Sara R Jaeger. 2015. Using Twitter data for food-related consumer research: A case study on "what people say when tweeting about different eating situations". *Food Quality and Preference* 45 (2015), 58–69.
- [75] Chong Wang and David M Blei. 2011. Collaborative topic modeling for recommending scientific articles. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*. 448–456.
- [76] Zied Zaier, Robert Godin, and Luc Faucher. 2008. Evaluating recommender systems. In *2008 International Conference on Automated Solutions for Cross Media Content and Multi-Channel Distribution*. IEEE, 211–217.
- [77] Brian D Ziebart, Andrew L Maas, J Andrew Bagnell, and Anind K Dey. 2009. Human Behavior Modeling with Maximum Entropy Inverse Optimal Control.. In *AAAI Spring Symposium: Human Behavior Modeling*, Vol. 92.